

Solution Requirements

Date: 11 June 2026

Project Name: E-commerce Application

Functional Requirements

1. User Management:

- The system must allow users to register, log in, and manage their profiles.
- The system must support Role-Based Access Control (RBAC) distinguishing between standard Users and Admins.

2. Product Catalog:

- The system must display a list of products with images, descriptions, prices, and stock levels.
- The system must allow users to search products by name and filter by category, price, and rating.

3. Shopping Cart & Checkout:

- The system must allow users to add items to a cart, update quantities, and remove items.
- The system must provide a multi-step checkout process (Shipping Address -> Payment Method -> Place Order).
- The system must integrate with Payment Gateway (Planned) to process credit card payments.

4. Order Management:

- The system must deduct stock inventory when an order is successfully placed.
- The system must allow users to view their order history and status.

5. Admin Operations:

- Admins must be able to perform CRUD operations on products, categories, and users.
- Admins must be able to update order fulfillment milestones (e.g., Shipped, Delivered).

Non-Functional Requirements

1. Security:

- Passwords must be hashed using Bcrypt before storage.
- API endpoints must be protected using JWT stored in HTTP-only cookies.
- The system must prevent unauthorized access to admin routes.

2. Performance:

- The application UI must load quickly, leveraging React's virtual DOM.

- API responses should be optimized; images must be served via Cloudinary CDN for fast loading.

3. **Usability:**

- The frontend must be fully responsive and provide a seamless experience on both desktop and mobile devices.

4. **Reliability:**

- Database transactions (NoSQL) must ensure order integrity during checkout to prevent overselling.